

Polylactic Acids for our Bioplastic Assets

Report 2

CHE 397 - Senior Design II

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Group 12

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Project Charter

Report 2

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Goal

Explain why producing PLA from sugarcane bagasse is economically feasible and environmentally friendly

Deliverables

- Final BFD and PFD
- Detailed Mass Balance
- Equipment List
- Equipment Sizing
- Preliminary Utilities

Design Plan

Task	Start Date	End Date	Assignee(s)
Logistics			
Tasks Assigned	02/11/2020	02/11/2020	All
Project Charter	02/11/2020	02/11/2020	All
Design Plan	02/11/2020	02/11/2020	All
Assignment			
Final PFD	02/11/2020	02/29/2020	All
Detailed Mass Balances	02/11/2020	02/29/2020	All
Equipment List	02/11/2020	02/29/2020	All
Equipment Sizing	02/11/2020	02/29/2020	All
Preliminary Utilities	02/11/2020	02/29/2020	All
Wrap-Up			
Proofreading & Editing	02/26/2020	02/29/2020	All & Mentor
Layout	02/11/2020	02/11/2020	All

Introduction

As the world population continues to increase, world food production is likely to increase as well. Therefore, the resources needed for producing the materials we use to package and contain, not just food but other commonly used materials, are likely to increase as well. Much of the packaging and containers currently in use are plastics, which are mainly produced with compounds derived from fossil fuels. However, the drawbacks of using such plastics include an increase in waste management (especially from single-use items), land and water pollution, and the release of CO₂ into the atmosphere from the crude oil refinery process.^[1] CO₂ is a greenhouse gas, which has been one of the main contributions to climate change. Because there is only a finite amount of available fossil fuels, producing plastics can become challenging if the current rate of global extraction does not decrease significantly. A study by Shogren, Wood, Orts, and Glenn notes that “Global plastics production in 2016 was 740 billion pounds with a growth rate of about 4% per year” and that the U.S. production alone “was 117 billion pounds in 2017.”^[1]

A solution to these problems could be to develop biobased plastics, which can act as a substitute to fossil fuel-based plastics. However, it is imperative that any biobased plastics in consideration are derived from plant material that does not require new land to be cultivated. This is because recent studies have concluded that cultivation of land to accommodate the growth of virgin feedstocks contribute greenhouse gas emissions. For instance, Escobar, Haddad, Borner, and Britz have shown that a “hypothetical 5% target for bioplastic consumption relative to total plastic demand in major producing regions is not an effective strategy for climate change mitigation if based on food crops.”^[2] Piemonte and Gironi have determined that when accounting for land-usage changes, “the net effect of the bioplastics production via clearing of carbon-rich habitats is to increase CO₂ emissions for decades or centuries relative to the emissions caused by petroleum-based plastics use.”^[3]

The biobased plastic in consideration is polylactic acid (PLA), which is a biodegradable polyester. PLA is produced by the polymerization of lactic acid, which is a product of the fermentation of sugars from carbohydrates. The benefit to using this process is that different plant-waste feedstocks that contain carbohydrates, such as sugarcane bagasse, can be used.^[1]

The objective of this project is to develop an economically viable production process for PLA pellets, which can be distributed to manufacturers, and perform a study estimate of the total

plant cost given a chosen location. The goal would be to optimize the purification process of lactic acid for PLA and promote a “circular economy”, where Shogren, Wood, Orts, and Glenn note “resource utilization and recycling/reuse is emphasized while minimizing the amount of waste produced.” [1]

Market Analysis

To effectively study the economic prosperity of producing PLA pellets from a feedstock of sugarcane bagasse, several factors must be considered. The key points that will be looked at are value chain, market drivers, and feedstock trends regarding the plant’s location in Sao Paulo, Brazil.

Value Chain

Polylactic acid is currently used in many kinds of commercial industries, therefore deeming it a successful market to work in. As of 2018, about 1,162 KT of bioplastics were produced, which included both non-biodegradable plastic and biodegradable plastic production. Non-biodegradable plastic production was at about 384 KT and was closely followed by starch-based plastics at 373 KT. Polylactic acid production sat in third at about 300 KT produced. The industries that utilize PLA range anywhere from water and beverage bottles, food service accessories, to processed food packaging and much more as seen in *Figure 1*.^[4]

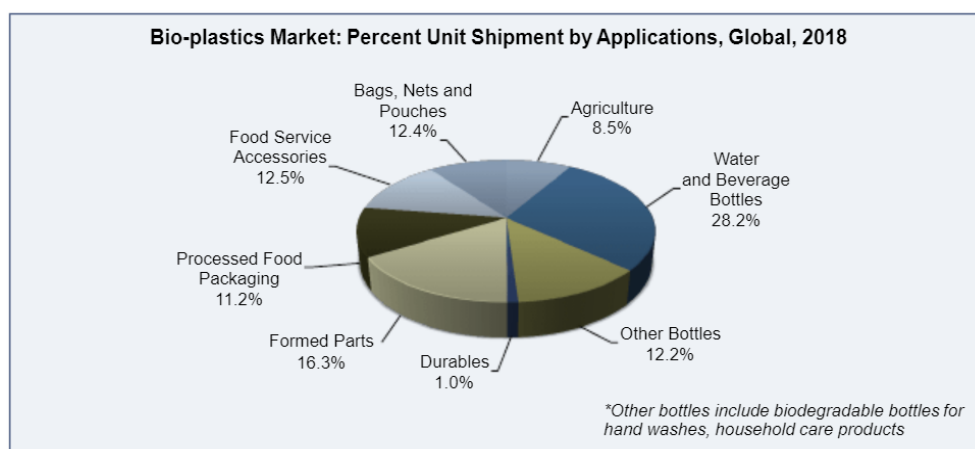


Figure 1: Bio-plastics Market Applications ^[4]

Market Drivers

The global market for PLA is projected to increase in the future as a result in the abundance of applications of bioplastics, as seen in *Figure 1*, as well as its ability to compete with the quality of traditional plastics made from crude-oil. Some of the great qualities this material carries are its transparency, stiffness, and glossy look as well as its ability to be a barrier over strong scents. Also, as the price of crude oil decreases and growth in the production capacities increases, the price of bioplastics will fall, making it more affordable. ^[5]

In general, from 2018 to 2023, bioplastic production is expected to increase by 11.8%. By the year 2023, as seen in *Figure 2*, PLA production is predicted to surpass its predecessors at about 693 KT produced, with non-biodegradable and starch-based plastics falling behind at 615 KT and 558 KT produced, respectively. The global revenue for bioplastics as of 2018 was \$5.32 billion, with a projected 9.5% growth by 2023, as seen in *Figure 3*.^[4] Since PLA makes up 25% of the bioplastic production, it would have generated a global revenue of \$1.3 billion in 2018. Analyzing these figures, it is clear that as the production of bioplastic increases, the prices will drop.

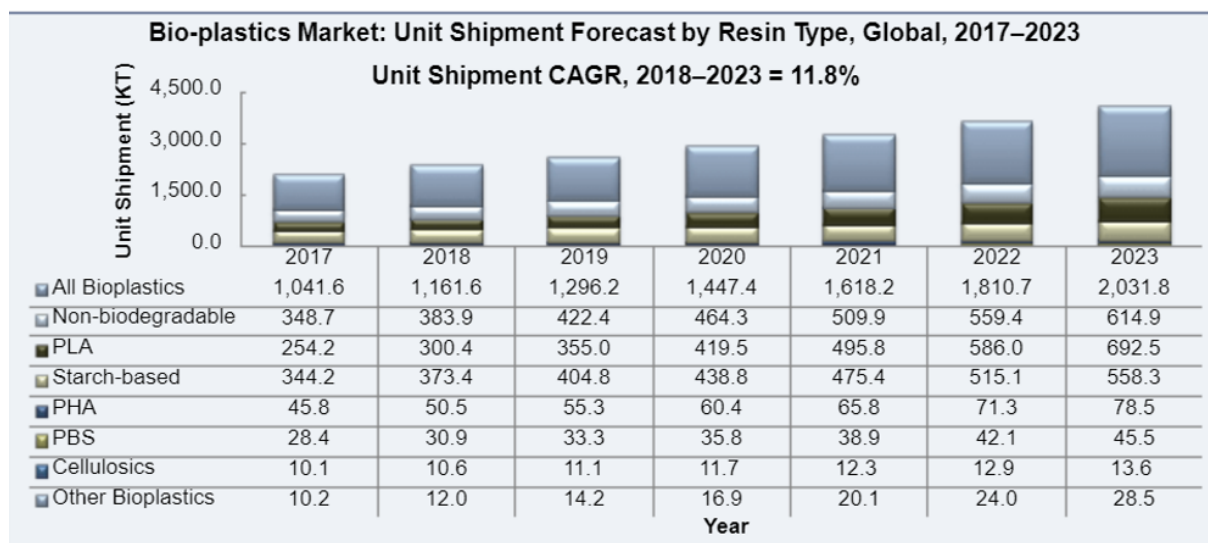


Figure 2: Bio-plastic Market Projections for Individual Resin Types ^[4]

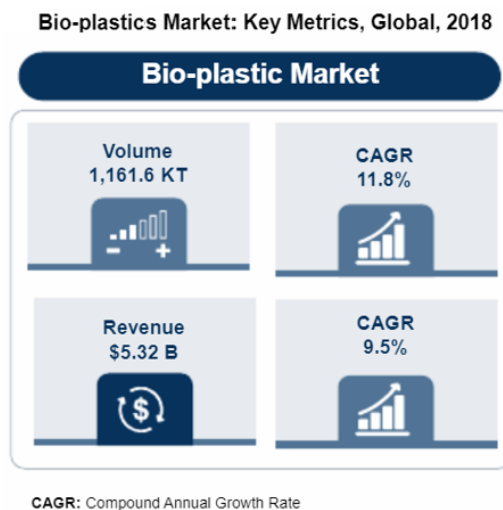


Figure 3: Bio-plastic Global Metrics ^[4]

In general, this processing plant's PLA will target the global market in order to make a profit. Upon examining *Figure 4*, it is seen that PLA import revenues are highest in Taiwan, the Netherlands, and Germany.^[6] Also seen in *Figure 5*, Asia and Australia will have a high growth rate in the next couple years. The rest of the world, except for Africa, will have moderate growth rates.^[7] The increased consumption of PLA in these regions is due to the public's consciousness about using sustainable and green packaging for their everyday products. Feedstock availability will be an important factor impacting PLA demand.^[6] PLA demand in these regions will continue to increase as higher demand for using PLA in food packaging as well as the other PLA end-uses previously shown in *Figure 1*.

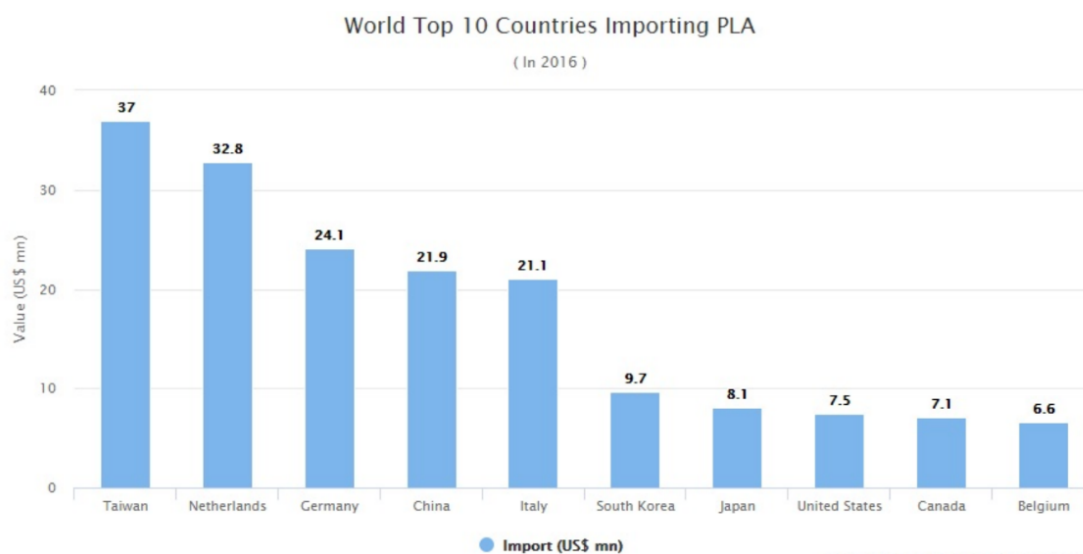


Figure 4: Top 10 Countries in PLA Consumer Market ^[6]

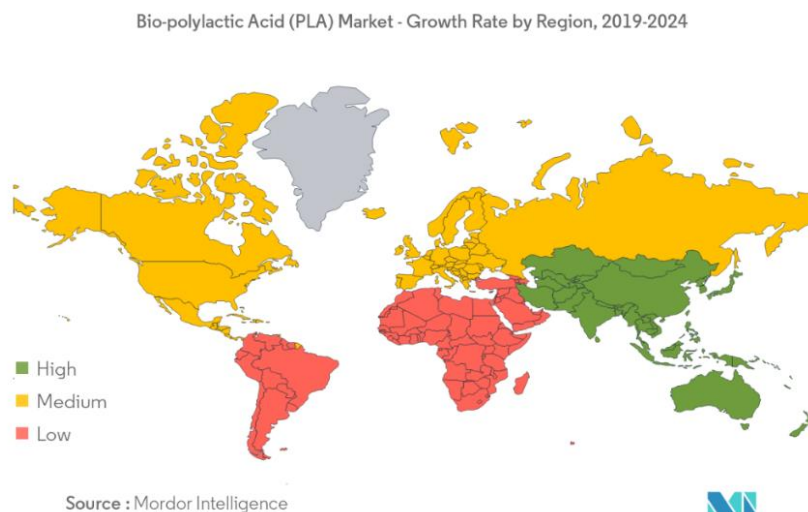


Figure 5: PLA Market Growth Rate ^[7]

Currently, the largest producer of bioplastic is the Asian-Pacific region (APAC) at a 55% production capacity as seen in *Figure 6*. The Asian-Pacific region is a larger contributor to the production capacity because of its low labor wages, cheap land, and readily available feedstocks.^[4] All these factors help boost this region's ability to compete globally with other plastics. Although the APAC region is the largest producer of bioplastics, the region itself doesn't consume a large number of bioplastics largely due to a stronger dependence on fossil fuel-based plastics.^[4] Following the APAC region is Europe and North America at 19% and 16% production capacity, respectively. The final 10% of the production capacity comes from the rest of the world, which is where the focus will shift to next.

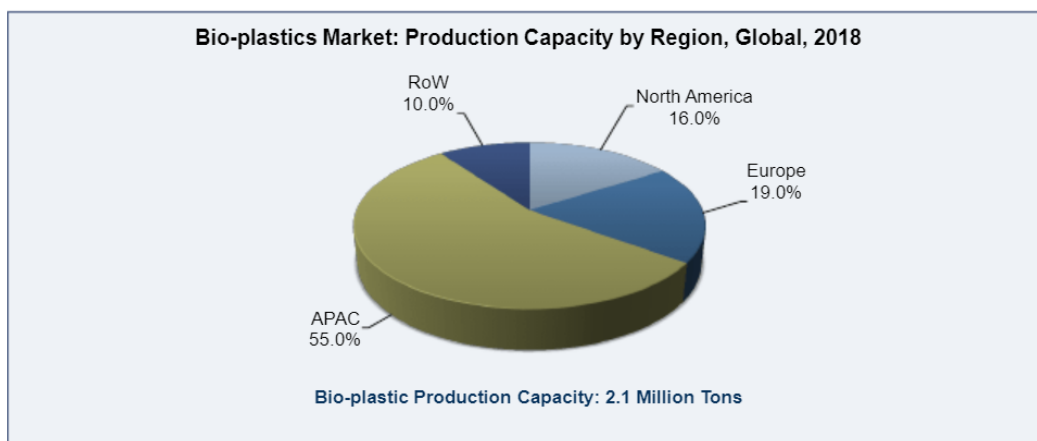


Figure 6: Bio-Plastic Production Capacity by Region ^[4]

Feedstock Trends

The feedstock for this plant is sugarcane bagasse. Brazil is the largest producer of sugarcane in the world.^[8] About 670 tons of sugarcane is produced there each year.^[9] Sao Paulo, a Brazilian municipality, specializes in growing commercialized sugarcane and is responsible for about 60% of the sugarcane grown in Brazil.^[10] Sao Paulo, will be the plant's location.

In addition to the large sugarcane production rate in Brazil, the consumer market for Brazilian sugarcane is also very large. As seen in *Figure 6*, this product is used a lot in order to make ethanol and sugars.^[11] Since sugarcane is widely used that would mean a reasonable amount of sugarcane bagasse is left behind. It is important to note that one ton of sugar cane equates to 250 – 270 kg of bagasse^[9] and costs about \$18 per ton of dry bagasse.^[12] The sugarcane bagasse also has many uses such as producing energy and biopolymers as seen in *Figure 7*.^[13] With that, sugarcane bagasse can fulfill its role in PLA production which makes it a sensible raw material competitor in Brazil.

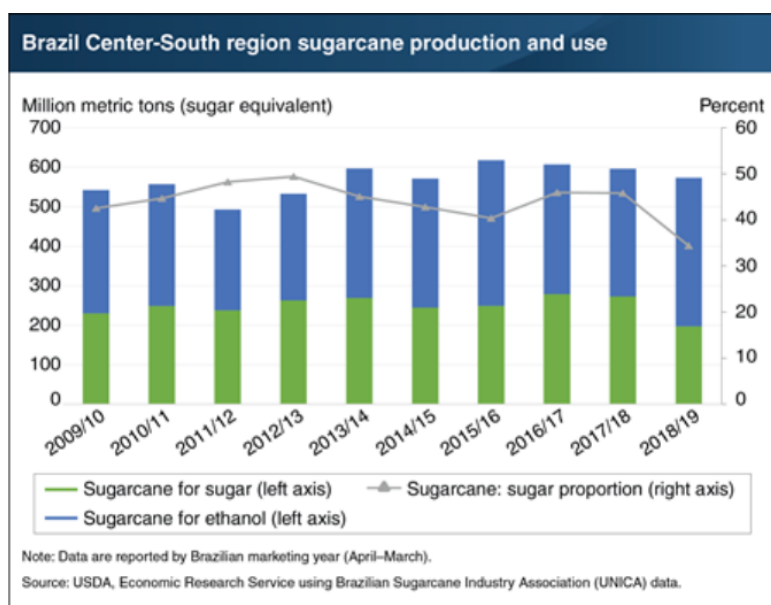


Figure 7: Brazilian Sugarcane Production and Use^[11]

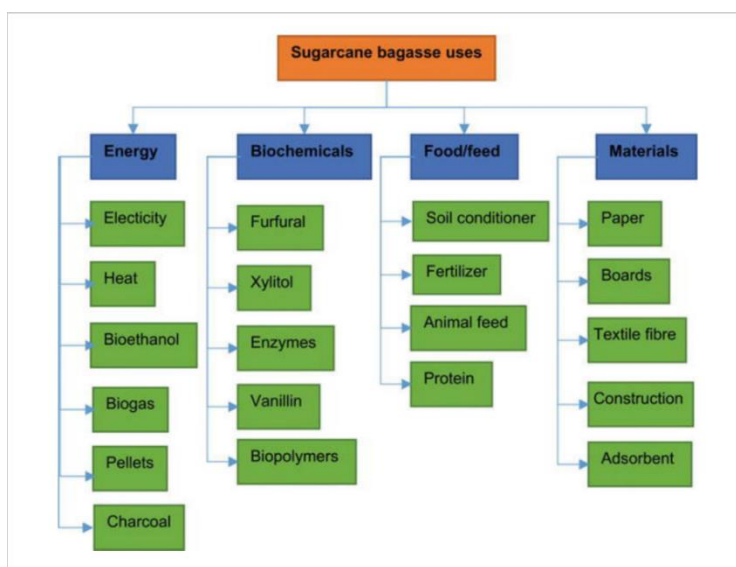


Figure 8: Sugarcane Bagasse Uses ^[13]

Process Description

Process

The production of polylactic acids occurs in a series of steps. A viable feedstock is subjected to a chemical reaction which allows sugars to be separated. Chemical synthesis or fermentation then takes place where the sugars are changed to lactic acid. There are many steps that come before the product becomes lactic acid, but that is dependent on the synthesis that takes place to combine the raw materials. The lactic acid is then purified and goes through a polymerization process into becoming polylactic acid, which can be used to make bioplastic products.

Possible Solutions

Possible solutions to produce polylactic acid involve the types of raw material to start with, methods of separating out sugars, changing of sugars into lactic acid, methods of separating and purifying lactic acid, and polymerizing lactic acid into PLA.

There are many raw materials which are cheap and have high concentrations of sugar that can be extracted to make lactic acids.^[14] Such raw materials include apple pomace, corncobs, potatoes, and sugarcane bagasse. These raw materials are waste products from other

processes, which can allow the production of lactic acid to be renewably sourced and promote a circular economy. Apple pomace has high amounts of free glucose and fructose which have the carbon sources needed for lactic acid production. There is also a high amount of polysaccharides to break down to give simple sugars as well as metal ions within the pomace that can reduce costs for extra nutrients needed for any fermentation process.^[15] Corncobs are readily available biomasses that contain high amounts of mixed sugars within them.^[16] The potato industry also has a lot of waste from its peels, which contain starches (carbohydrates) that can be used to produce lactic acid. This solution is considered to be eco-friendly because the starch can be converted into lactic acid without any emissions of CO₂.^[17] The benefit of using sugarcane bagasse as a raw material is that it is a byproduct of sugar production.^[8] The sugarcane bagasse contains high amounts of lignocellulosic materials, which contains high amounts of fermentable sugars.^[9] Table 1, shown below, shows the productivity and yield of some of these raw materials to create lactic acid. Along with the productivity and the yield, the amount of raw materials that are being produced must also be taken into consideration.

Table 1: Comparison of Lactic Acid Production Using Various Raw Materials ^[16]

Substrate	Organism	Fermentation process	Lactic acid			References
			Concentration (g l ⁻¹)	Productivity ^a (g l ⁻¹ h ⁻¹)	Yield ^b (g g ⁻¹)	
Alfalfa fibers	<i>Lactobacillus delbrueckii</i>	SSF	35.4 ^e	0.75	0.35	Sreenath et al. (2001)
	<i>Lactobacillus plantarum</i>	SSF	46.4 ^e	0.64	0.46	Sreenath et al. (2001)
Apple pomace	<i>Lactobacillus rhamnosus</i> CECT-288	Batch	32.5 ^c	5.4	0.88	Gullon et al. (2008)
Barley bran hydrolysates	<i>Lactobacillus pentosus</i>	Batch	33.0 ^e	0.60	0.57	Moldes et al. (2006)
Cellobiose and cellotriose	<i>Lactobacillus delbrueckii</i> Uc-3	Batch	90.0 ^e	2.3	0.90	Adsul et al. (2007)
Cellulose	<i>Lactobacillus delbrueckii</i> NRRL-B445	SSF	65.0 ^e	0.18	–	Iyer and Lee (1999)
Corncob	<i>Acromonium cellulose</i> and <i>Rhizopus</i> sp.	SSF	24.0 ^c	0.17	0.24	Miura et al. (2004)
	<i>Lactobacillus pentosus</i>	Batch	26.0 ^c	0.34	0.53	Moldes et al. (2006)
Lignocellulosic hydrolysates	<i>Lactobacillus</i> sp. RKY2	Cell-recycle	27.0 ^e	6.7	0.90	Wee and Ryu (2009)
Molasses	<i>Lactobacillus delbrueckii</i> mutant Uc-3	Batch	166 ^e	4.2	0.87	Dumbrepatil et al. (2008)
	<i>Lactobacillus rhamnosus</i> ATCC 7469	SSF	73.0 ^e	2.9	0.97	Marques et al. (2008)
Paper sludge	<i>Bacillus coagulans</i> strains 36D1	SSCF	92.0 ^e	0.96	0.77	Budhavaram and Fan (2009)
	<i>Bacillus coagulans</i> strains P4–102B	SSCF	91.7 ^e	0.82	0.78	Budhavaram and Fan (2009)
Rice bran	<i>Lactobacillus delbrueckii</i> IFO 3202	SSF	28.0 ^d	0.78	0.28	Tanaka et al. (2006)
Sugar cane bagasse	<i>Bacillus</i> sp.	Batch	55.5 ^c	0.39	0.77	Patel et al. (2004)
	<i>Lactococcus lactis</i> IO-1	Batch	10.9 ^c	0.17	0.36	Laopaiboon et al. (2010)
Trimming vine shoots	<i>Lactobacillus pentosus</i>	Batch	24.0 ^e	0.51	0.76	Moldes et al. (2006)
Wastepaper	<i>Lactobacillus delbrueckii</i> B445	SSF	31.0 ^e	–	–	Schmidt and Padukone (1997)
Wheat bran hydrolysate	<i>Lactobacillus bifementans</i>	Batch	62.8 ^e	1.2	0.83	Givry et al. (2008)
Wood hydrolysate	<i>Enterococcus faecalis</i> RKY1	Batch	93.0 ^c	1.7	0.93	Wee et al. (2004)
Xylose	Recombinant <i>Lactobacillus plantarum</i>	Batch	41.2 ^d	–	0.82	Okano et al. (2009b)
Corncob molasses	<i>Bacillus</i> sp. strain	Fed batch	74.7 ^c	0.38	0.50	Present study

In general, lactic acids contain L-lactic, D-lactic isomers, or DL-lactic acids ^[16] in which the production of polylactic acids depends on. Pure isomers are better for production than the mixture of DL-isomers. These isomers are dependent on the production process of lactic acid, which can be either through chemical synthesis or fermentation.

The chemical synthesis process produces the lactic acid by using chemicals that are taken from petroleum. The chemical synthesis occurs by hydrogen cyanide being added to the acetaldehyde to make lactonitrile. This is then hydrolyzed by hydrochloric acid or sulfuric acid to produce the lactic acid along with another byproduct. Then, an esterification process and a hydrolysis process with water occurs to produce the lactic acid.^[18] This process makes a mixture with DL-lactic isomers. Another downside to the chemical synthesis is that the process isn't "economically and technically feasible".^[19]

Fermentation only processes raw materials with high amounts of sugars, which can be taken from the raw materials that have been mentioned before. There are microorganisms that are used during this process to help create the lactic acid; they are categorized as bacteria, fungi, and yeast. The microorganisms are a key factor in deciding the yield, productivity, and purity of the lactic acid. Out of the three categories, bacteria are the type of microorganisms used to create the lactic acid commercially.^[19] There are two types of fermentation processes used with the bacteria, which are homofermentative or heterofermentative.^[20]

In the homofermentative process, the bacteria converts the glucose from the raw material into lactic acid with a conversion of 85%.^[18] It also decreases the number of by-products being produced which is why it is in demand for commercial use.^[14] This process also allows there to be pure isomers of L-lactic and D-lactic acids. In the heterofermentative process, glucose is converted into lactic acid with a conversion of 50% along with CO₂ and ethanol. The ratio between the glucose, lactic acid, CO₂, and the ethanol is a one to one mol ratio.^[20] This type of process also produces the DL-lactic acid, which is not desirable.

There are three conventional fermentation methods that can be considered: batch fermentation, fed-batch fermentation, and continuous fermentation. In batch fermentation, all the components needed for the process are added in together before the process begins. It is a closed system, so it prevents contaminants from coming in and allows there to be a production of high concentrated lactic acids. There are two types of batch fermentation: solid-state fermentation (SSF) and separate hydrolysis and fermentation (SHF). In the solid-state fermentation, water is neglected in the process and there's lower levels of enzymes that are loaded into the method. The process time for this type of fermentation is quick which is an advantage over SHF. In the separate hydrolysis and fermentation process, there is a pretreatment of the raw materials to get

rid of impurities. The pre-treatment causes a decrease in the levels of productivity, which reduces the yield of the lactic acid.^[19]

In the fed-batch fermentation process, all the components are added into the fermentation solution at a constant rate and at constant time intervals.^[19] The benefit of the fed-batch fermentation is that it produces higher concentrations of lactic acids but must be monitored, since the build-up of inhibiting substances can occur after multiple feed cycles. This can lower productivity. In the continuous fermentation, a fresh feed is constantly added into the fermenter while the previously fermented solution is taken out.^[19] It has reported that the productivity levels are two times higher than they were in the batch and the fed-batch fermenters since shorter residence times are achieved.^[21] However, the operation of continuous fermentation is costly compared to the other two processes.^[19]

After the fermentation step, the lactic acid needs to go through a polymerization process to produce the polylactic acid. To do this, the lactic acid must be purified first. There are different ways to purify lactic acid, which include electrodialysis, adsorption, reactive extraction, and esterification/reactive distillation. Electrodialysis is a process that separates components through a selective membrane by a DC electric force. This process maintains a constant pH level in the fermented solution while continuously removing the lactic acid from this solution. The cons from this process is that there must be pretreatment in order to remove impurities and effects of polarization when this process is done at a large scale.^[20]

Adsorption is a process that uses solid sorbents throughout the process of creating lactic acid to increase yield. The sorbents have high selectivity to lactic acid and can replace the lost lactic acids in the batch. The downside to this process is that it needs to have the resins restored along with corrections in the pH of the feed at a large-scale process. Reactive extraction is a process where the lactic acid can be removed from the fermented solution through reaction with an extractant and separation through a stripper. Caution needs to be considered with the use of the extractant to prevent the fermented solution from becoming hazardous. The final type of process is esterification and reactive distillation in which there is a distillation process occurring simultaneously to the process of esterification. Before the process, there must be a pretreatment to remove the contaminants in the lactic acid.^[20]

After the purification process, the lactic acid is polymerized into polylactic acids. Possible polymerization procedures include lactic condensation and coupling, azeotropic dehydration and condensation, and Ring Opening Polymerization (ROP) of Lactide. The objective of this step is to increase the molecular weight of the polylactic acid while keeping the purity level high. The lactic condensation and coupling process involves the use of coupling agents to increase the molecular weight of the polylactic acid that is produced through the condensation polymerization. The coupling agents react with the hydroxyl group to create the polymer. There may be impurities in the final product from the catalysts, where some of these agents could cause the product to be less biodegradable. Azeotropic dehydration and condensation requires the lactic acid and the catalyst to be azeotropically dehydrated to obtain the higher values in the molecular weight of the polylactic acid. The downside to this process is that there are residues from the catalyst that can cause degradation or hydrolysis to occur. Also, there can be toxicity due to the catalyst in the application, which can be harmful to the consumers.^[14]

The ROP of lactide method is the only type that makes pure polylactic acids with high molecular weights.^[14] There are two things that occur in this method, where one is to form the lactide and the other is the ring opening polymerization. The lactic acid is dehydrated to produce the lactide ^[20] and then the ROP process is done in bulk with a low level of a catalyst that is non-toxic. Usually tin compounded catalysts are used because of their “high catalytic efficiency, low toxicity, food and drug contact approval, and ability to give high molecular weights with low racemization.”^[14] Also, ROP is favored due to the control and quality in the production process.^[22]

Chosen Solution

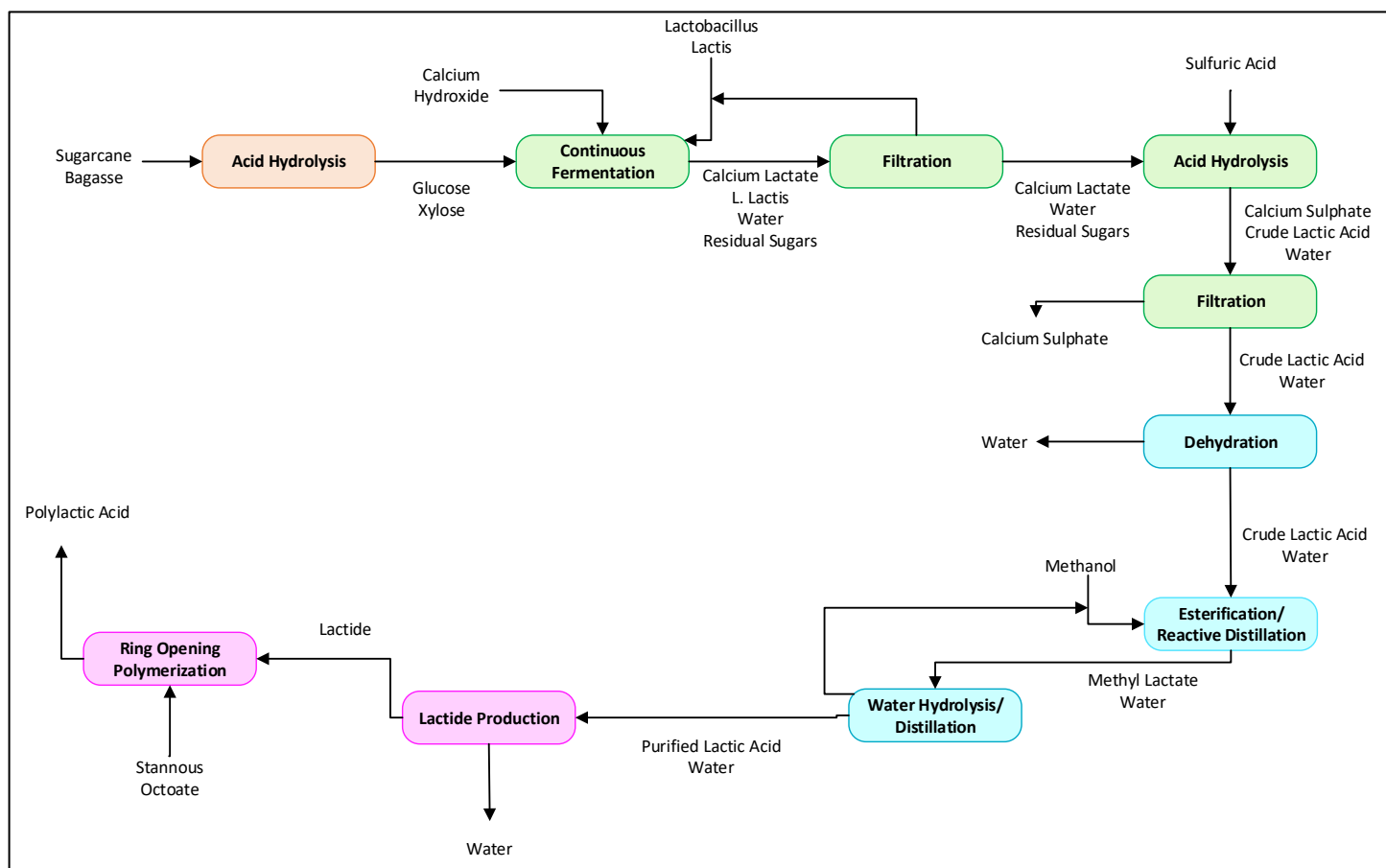


Figure 9: BFD of the Polylactic Acid Production from Sugarcane Bagasse

The chosen solution for this project is to use sugarcane bagasse as the raw material which will undergo a process of staggered-continuous fermentation. This process will be homofermentative and will utilize *Lactobacillus Lactis* bacteria as the microorganism. After the fermentation process, the products will go through a series of steps to get purified lactic acid. These steps involve filtration, acid hydrolysis, esterification distillation, and water hydrolysis distillation. The purified lactic acid will be subjected to dehydration, where lactide is formed. This lactide will be polymerized using ROP, which produces polylactic acid and will be formed as resins to be used as a raw material for bioplastic manufacturers. The overall process is demonstrated in *Figure 9*, which is the BFD of the chosen process.

Sugarcane is the raw material that is chosen for this process. In *Table 1*, apple pomace had a higher yield and productivity than sugarcane bagasse, but the availability of the apple pomace is lower than sugarcane bagasse. Apple pomace is produced in millions of tons

worldwide while there are countries producing close to billion tons of sugarcane bagasse. The sugarcane bagasse goes through an acid hydrolysis step in order to extract the fermentable sugars that will be used in the fermentation process. In the acid hydrolysis, the type of acid being used will be the sulfuric acid. In *Table 2*, it is shown that 1.25% H_2SO_4 extracted higher amounts of sugar at a lower time, which is needed in the fermentation process. The acid hydrolysis will increase the amount of fermentable sugars to 34%.^[23] A simulation of this step will not be made with Aspen Plus. However, an assumption will be made where all of the lignin contained in the bagasse may be removed after acid hydrolysis via centrifuge as shown in *appendix III*.

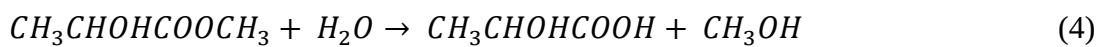
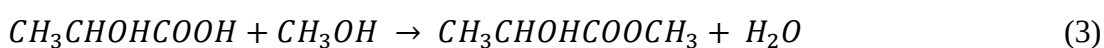
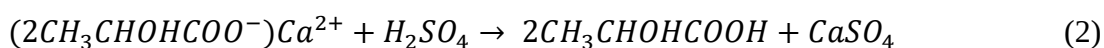
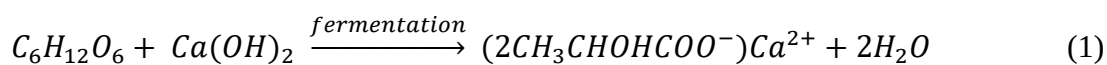
Table 2: Comparison of the Main components from the Acid Hydrolysis Treatment^[8]

Acid hydrolysis conditions (references)	Xylose (g l ⁻¹)	Glucose (g l ⁻¹)	Arabinose (g l ⁻¹)	Total sugar (g l ⁻¹)	Acetic acid (g l ⁻¹)	Furfural (g l ⁻¹)	E
0.5% of HCl, 100 °C, 5 h (this work)	22.59	1.50	1.29	25.38	0.15	1.19	10.85
2% of H_2SO_4 , 122 °C, 24.1 min (Aguilar et al. (2002))	21.6	3.00	ND*	24.60	3.65	0.52	4.76
6% of HNO_3 , 122 °C, 9.3 min (Rodríguez-Chong et al. (2004))	18.60	2.87	2.04	23.51	0.90	1.32	7.30
4% of H_3PO_4 , 122 °C, 5 h (Gámez et al. (2006))	17.60	3.00	2.60	23.20	4.00	1.20	3.74
2.5% of HCl, 140 °C, 30 min (Chandel et al. (2007))	21.50	5.84	2.95	30.29	5.45	ND	4.70
1.25% of H_2SO_4 , 121 °C, 2 min (Cheng et al. (2008))	17.10	7.20	2.00	26.30	4.00	1.40	4.11

The process of continuous fermentation using *Lactobacillus lactis* is chosen due to the result of high yields of L-lactic acid.^[21] The production of L-lactic acid is critical due to D-lactic acid being harmful to humans.^[20] Since, the PLA resins are optimized with a high purity, it is expected to be used in different industries. L-lactic acid is preferred in food and pharmaceutical industries due to humans only being able to metabolize L-lactic acids.^[20] During this fermentation process, the extracted fermentable sugars and *Lactobacillus lactis* is added along with calcium hydroxide. The calcium hydroxide is added to neutralize the acid in the fermentation solution as well as to create lactate products. Multiple fermenters will be used as shown in *appendix III* and will operate in a staggered configuration such that maintenance can be performed on one while the others are producing calcium lactate. The fermentation process has operating conditions of a temperature around 40°C, along with low oxygen levels, and an acid solution with a pH of approximately 6.^[14] The chosen bacteria allow there to be a conversion of 85% into the wanted product.^[20] The fermentation process will be simulated using Aspen Plus and with a stoichiometric reaction, since the fermentation kinetics were unable to be obtained. The product made from the fermentation step are calcium lactate along with residual sugars and residual cells. These products must be filtered out to get the calcium lactate. A centrifuge will be

used to separate out any bacteria left in the product stream and sent to a chemostat, which keeps the bacteria's nutrients constant allowing it to re-enter the fermentation unit again. The sizing of both centrifuges (CF-101 and CF-102) were modeled with the appropriate flowrates of their respective input streams, as shown in *appendix III*, along with calcium sulfate acting as the solid since lignin and lactobacillus lactis are unavailable components in Aspen Plus. The filtered calcium lactate solution will be treated with acid hydrolysis using sulfuric acid. This allows the calcium to break from the molecule to form crude lactic acid and calcium sulphate. The conversion rate of lactic acid yielded from this hydrolysis step is 97% ^[24] and will have calcium sulfate separated through filtration. An assumption needs to be made where most of the calcium hydroxide and sulfuric acid solution will react to yield very low concentrations of both compounds in their respective separation units. The crude lactic acid goes through an esterification step where methanol is reacted with lactic acid to form methyl lactate and water. The conversion of the methyl lactate is 85%.^[18] The methyl lactate solution goes through the reactive distillation, where the temperature is at reflux temperature. This process takes around 4-8 hours.^[20] From this distillation, methyl lactate is distilled from the water and any unreacted methanol. 90% of the methyl lactate is removed from the water and the unreacted methanol. Then, the methyl lactate will undergo a hydrolysis using water which will react to create purified lactic acid and methanol as the byproduct.^[18] 97% of the methanol will be removed from this process.^[8] It should be noted that the esterification distillation and water hydrolysis distillation will be simulated in Aspen Plus as separate reactor and distillation units as shown in *appendix III*. This method will be able to yield rigorous distillation results and simplify the overall simulation.

The following equations show the chemistry behind the fermentation process until the purification of lactic acid.^[18]



Equation 1 shows the chemical reactions during the fermentation and filtration step. *Equation 2* shows the acid hydrolysis step using H_2SO_4 . *Equation 3* shows the esterification step and *Equation 4* shows the water hydrolysis step.^[18]

After the lactic acid has become purified, it will go through the ROP of lactide process. This process is chosen due to the polymerization occurring with this process yields into having PLA with higher molecular weights.^[20] In this process, the purified lactic acid will undergo a dehydration process and lactide will be formed. The lactide will then undergo the ROP process using the coordination insertion method due to it increasing the molecular weight of the PLA being produced as well as reducing the amount of impurities. The ROP process will not be simulated with Aspen Plus. This is because polylactic acid is not a component in the available databanks and not all of the kinetics were able to be found. The polymerization will undergo an extrusion process which will complete within 5-7mins.^[14] This produces the final PLA product.

In conclusion, this method will be utilized for the basis of production of polylactic acid in the processing plant. Through the chosen solution, the yield for the molecular weight of the polylactic acids will be high. Also, through the optimization of the purification process of lactic acid, higher purities of polylactic acid will be obtained.

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Appendix I: Design Basis

The plant's location will be in Sao Paulo, Brazil because of its high production rates of the raw material needed, sugarcane bagasse. The purity of the product will be at 99%. The plant will operate for 333.33 days of the year for 24 hours equating to 8000 hours per year. As previously mentioned 300KT/year of PLA is produced globally and will be used in order to determine the plant's production rate. An assumption is made that the production rate will be equivalent to 10% of the global demand meaning this plant's goal is to produce 30 KT/year of PLA. In order to produce this amount of product, 152,926 kg/hr of raw material will be used.

Table 3: Design Basis of Processing Plant

Plant Location	Sao Paulo, Brazil
Raw Material	Sugarcane Bagasse
Product Material	Polylactic Acid
Amount of Raw Material	152926 kg/hr
Product Purity	99%
Hours of Operation	8000 hrs.
Plant Production Rate	30 KT/year

Appendix II: Block Flow Diagram (BFD)

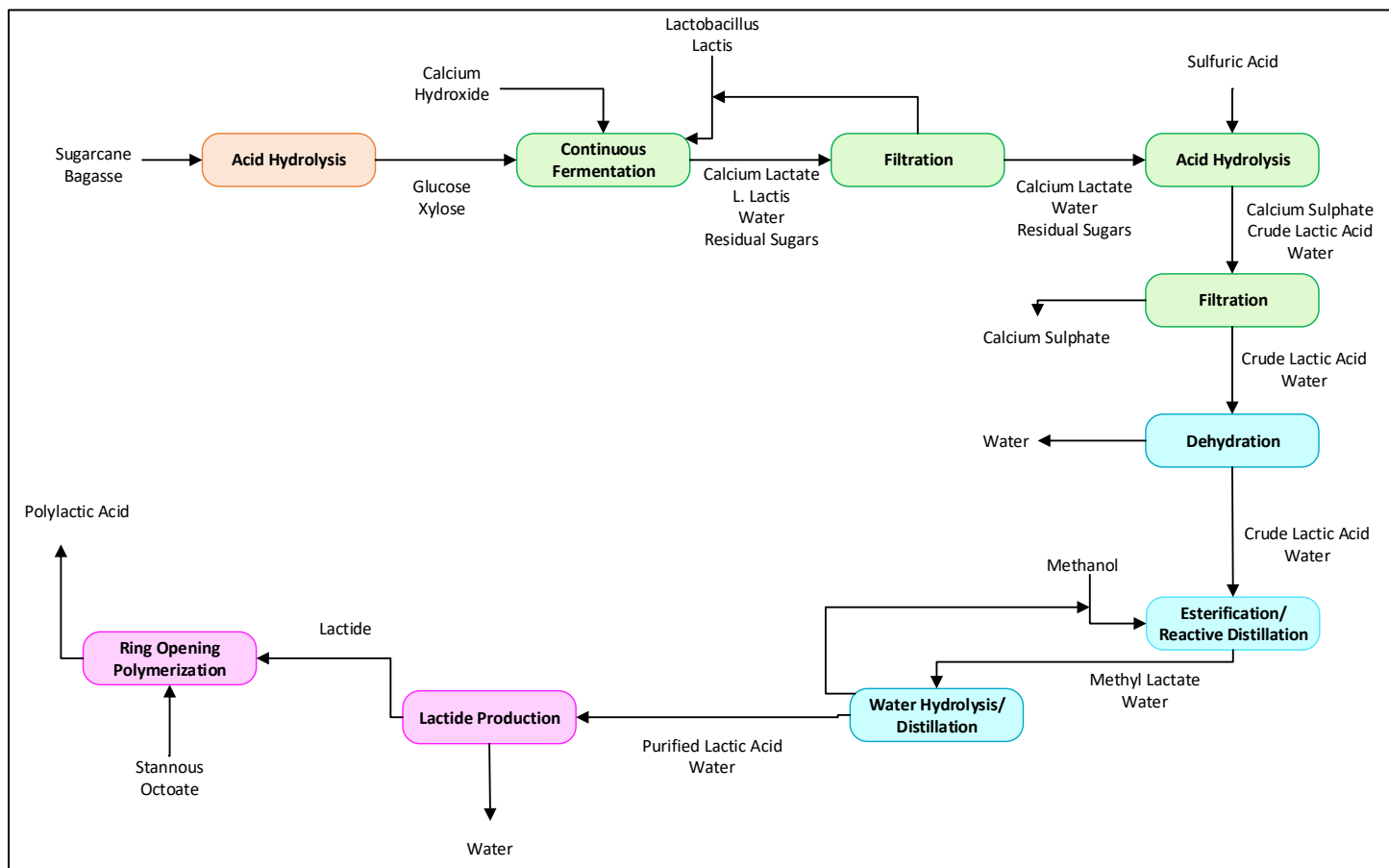


Figure 10: BFD for the Polylactic Acid Process from Sugarcane Bagasse

Appendix III: Process Flow Diagram (PFD)

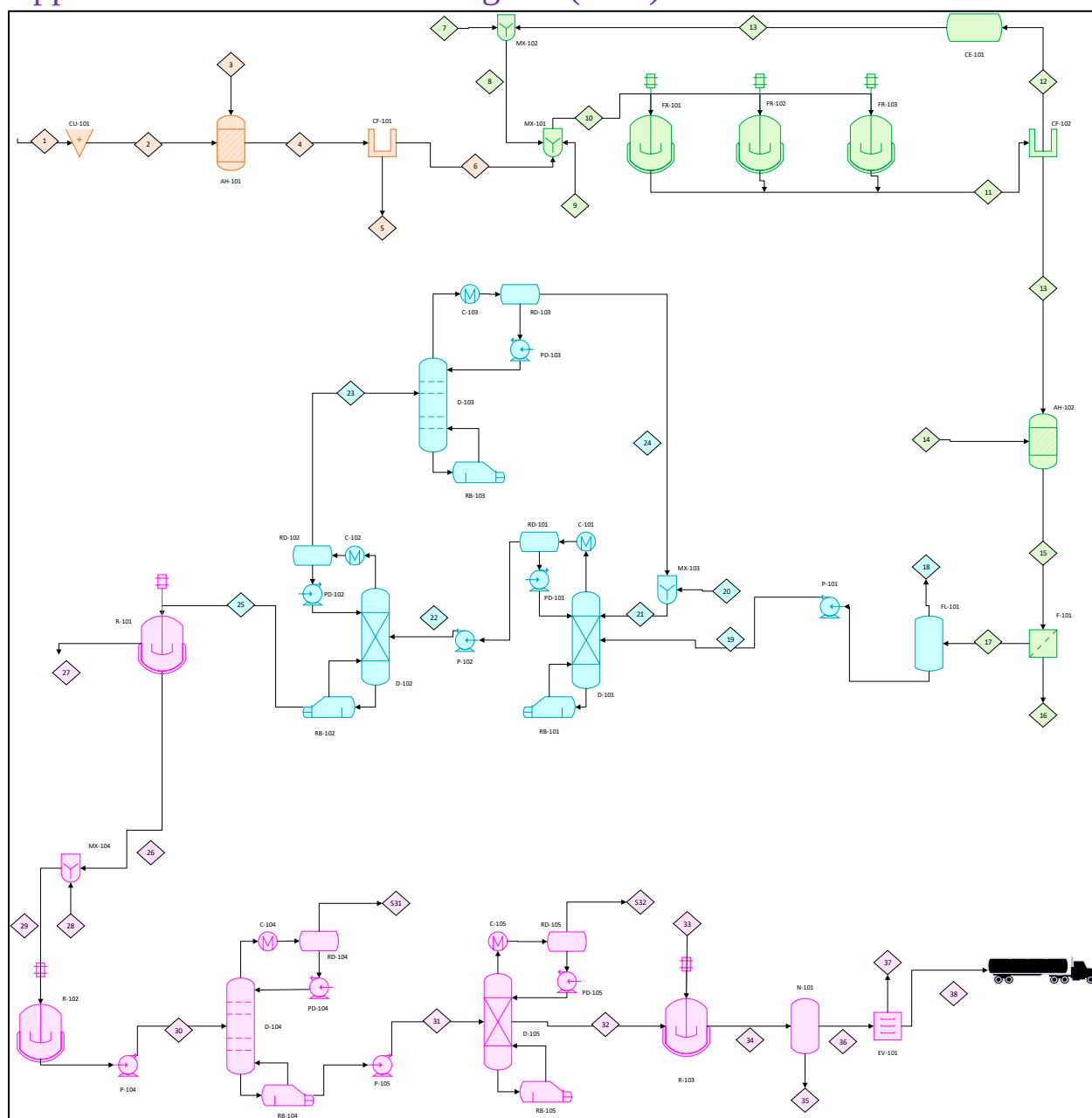


Figure 11: PFD of the Polylactic Acid Process from Sugarcane Bagasse

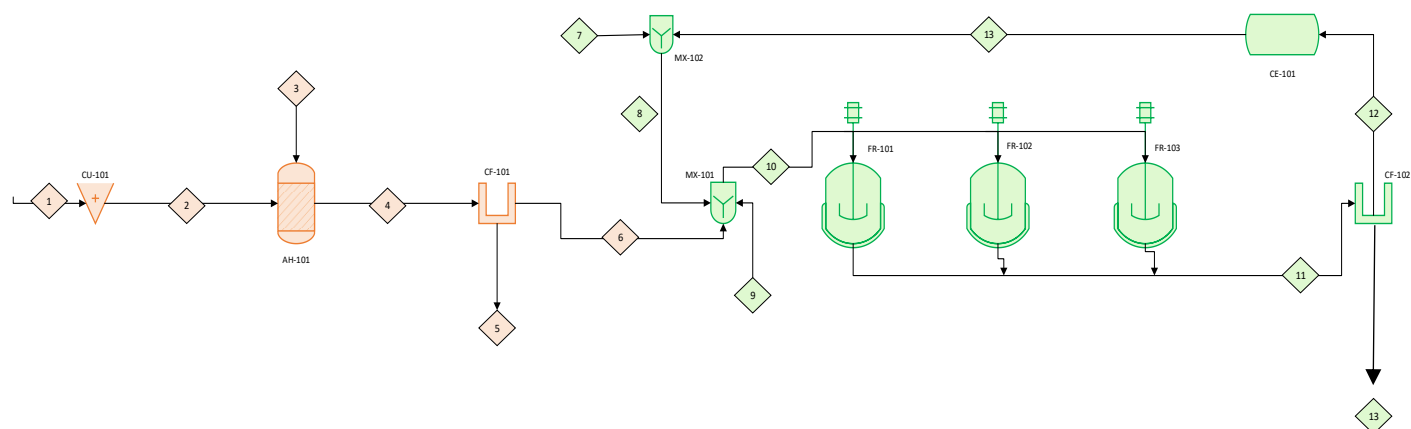


Figure 12: Pretreatment and Fermentation of Lactic Acid Process

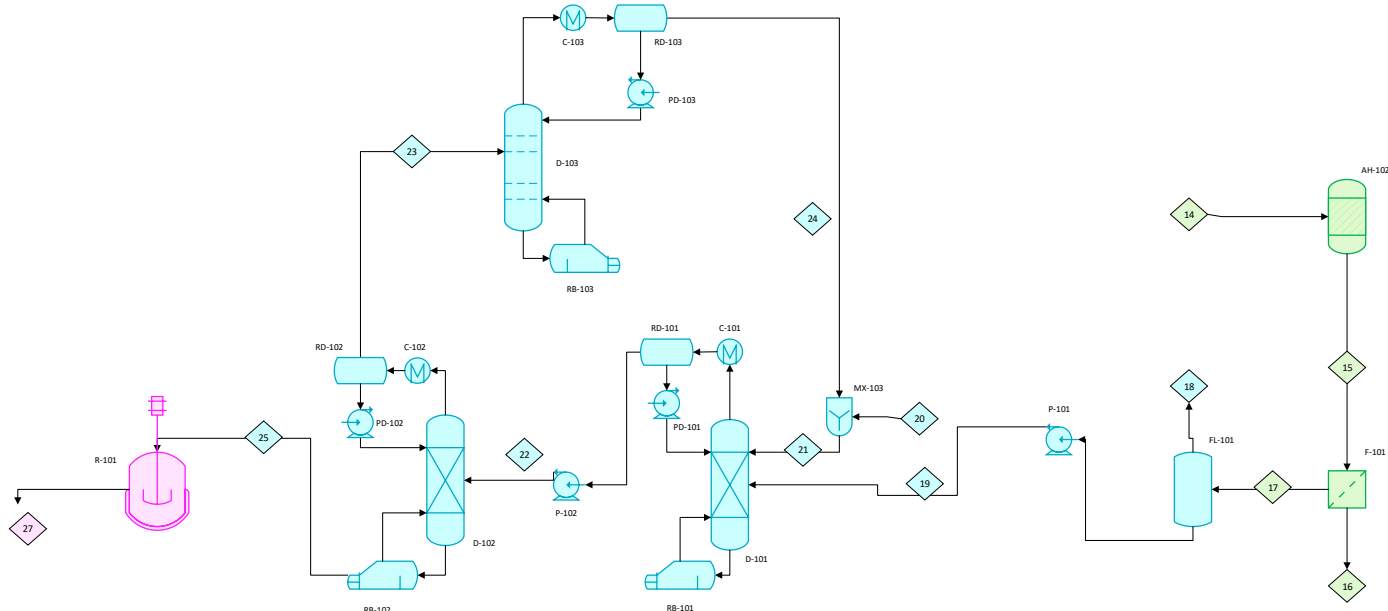
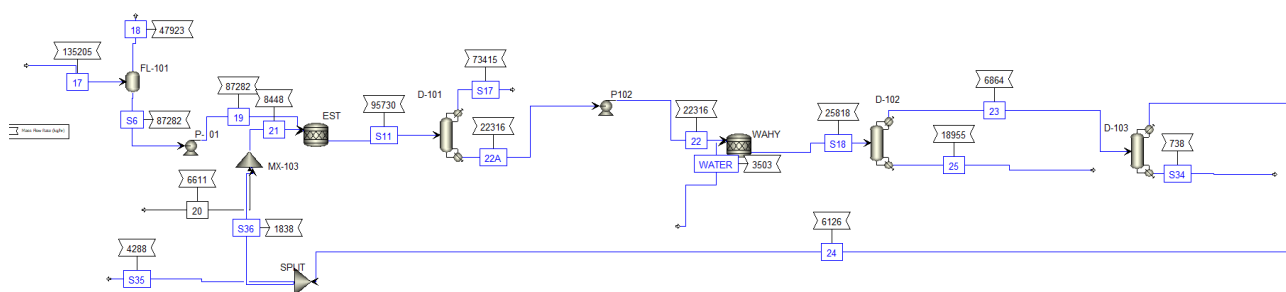
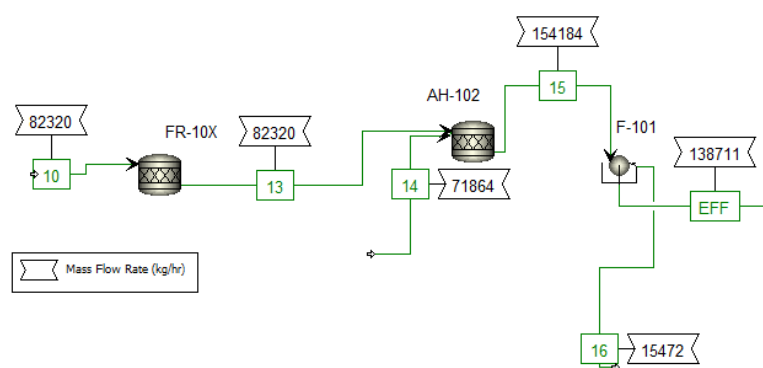
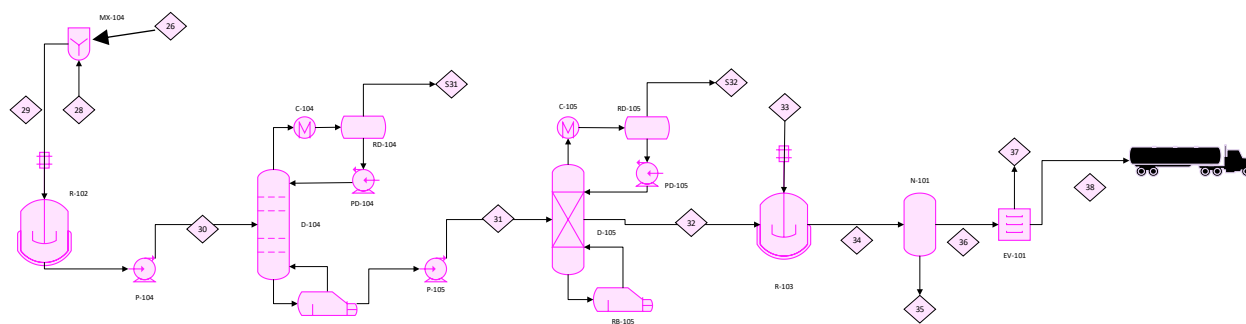


Figure 13: Fermentation into Purification of the Lactic Acid Process and Beginning of Polymerization Process



Appendix IV: Detailed Mass Balance

Table 4: Detailed Mass Balance for Pretreatment Process (Hand Calculated)

		Units	Stream			
Properties			2	4	5	6
	Temperature	K	298	368	298	298
	Pressure	atm	1	1	1	1
	Total Flow Rate	kg/hr	152926	152926	20046	162080
	Phase		Solid	Solid/Liquid	Solid	Liquid
Components	Cellulose	kg/hr	18400	0	0	18400
	Hemicellulose	kg/hr	10800	0	0	10800
	Lignin	kg/hr	9200	9200	9200	0
	Ash	kg/hr	1600	1600	1600	0
	Sulfuric Acid	kg/hr	2045.9	2045.9	2045.9	0
	Water	kg/hr	110880	110880	0	110880
	Glucane	kg/hr	0	5600	5600	0
	Glucose	kg/hr	0	12800	0	12800
	Xylane	kg/hr	0	856	856	0
	Xylose	kg/hr	0	8000	0	8000
	Arabinan	kg/hr	0	744	744	0
	Arabinose	kg/hr	0	1200	0	1200

Table 5: Detailed Mass Balance for the Fermentation Process

		Units	Stream					
Properties			10	13	14	15	16	17
	Temperature	K	298	298	293	368	293	298
	Pressure	atm	1	1	1	1	1	1
	Total Flow Rate	kg/hr	82320	82320	71864	154184	15472	135205
	Phase		Liquid	Liquid	Liquid	Liquid/Solid	Liquid/Solid	Liquid
Components	Glucose	kg/hr	22000	2200.0	0	2200	22.0	0
	Calcium Hydroxide	kg/hr	9045	905	0	905	9.05	100.96
	Calcium Lactate	kg/hr	0	23983	0	1199	12.0	0
	Water	kg/hr	51272	55232	61084	116316	1163	116253
	Sulfuric Acid	kg/hr	0	0	10780	539	5.39	50.478
	Lactic Acid	kg/hr	0	0	0	18811	188	18800
	Calcium Sulphate	kg/hr	0	0	0	14215	14073	0

Table 6: Detailed Mass Balance of Purification Process

		Units	Stream								
Properties			17	18	19	20	21	22	23	24	25
	Temperature	K	298	375	375	303	338	413	349	339	484
	Pressure	atm	1	1	2	2	1	1.3	1.2	1	1.2
	Total Flow Rate	kg/hr	135205	47923	87282	6610.5	8448.2	22316	6863.7	6125.5	18955
	Phase		Liquid	Vapor	Liquid	Liquid	Liquid/Vapor	Liquid	Vapor	Vapor	Liquid
Components	Calcium Hydroxide	kg/hr	100.96	0	100.96	0	0	100.96	0	0	100.96
	Water	kg/hr	116253	47876	68377	0	34.311	783.73	640.89	114.37	143.27
	Sulfuric Acid	kg/hr	50.478	3.59E-03	50.474	0	0	50.474	0	0	50.474
	Lactic Acid	kg/hr	0	46.897	18753	0	0	937.67	0	0	18449
	Methanol	kg/hr	0	0	0	6610.5	8413.9	0	6222.8	6011.1	6.3149
	Methyl Lactate	kg/hr	0	0	0	0	0	20443	2.34E-02	0	204.41

Table 7: Detailed Mass Balance for Polymerization Process (Hand Calculated)

		Units	Stream							
Properties			25	26	27	28	29	30	31	S31
	Temperature	K	419	430	273	430	443.15	450	450	330
	Pressure	atm	1.2	1.2	1.2	1.2	1.2	1.2	1	1.2
	Total Flow Rate	kg/hr	18955	18457	134.99	6	18457	18457	18446	10.036
	Phase		Liquid	Liquid	Liquid	Solid	Solid/Liquid	Liquid	Liquid	Vapor
Components	Water	kg/hr	143.27	8.2810	134.99	0	8.2810	8.2810	0	7.4529
	Lactic Acid	kg/hr	18449	258.29	0	0	258.29	25.829	23.246	2.5829
	Lactide	kg/hr	0	0	0	0	0	232.46	232.46	0
	Poly(lactic Acid)	kg/hr	0	18191	0	0	18191	18191	18191	0
	Catalyst	kg/hr	0	0	0	6	0	0	0	0
	Stream									
	Water	kg/hr	32	S32	33	34	35	36	37	38
	Lactic Acid	kg/hr	360	330	320	330	330	330	360	300
	Lactide	kg/hr	1	1.2	1	1.2	1.2	1	1.2	1
	Poly(lactic Acid)	kg/hr	18400	69.737	6	18625	6	18625	493.85	18006
	Catalyst	kg/hr	Liquid	Vapor	Solid	Solid/Liquid	Solid	Solid/Liquid	Vapor	Solid
	Water	kg/hr	0	0	0	0	0	0	493.85	0
	Lactic Acid	kg/hr	0	23.246	0	0	0	0	0	0
	Lactide	kg/hr	209.21	46.491	0	209.21	0	209.21	0	83.685
	Poly(lactic Acid)	kg/hr	18191	0	0	18416	0	18416	0	17922
	Catalyst	kg/hr	0	0	6	0	6	0	0	0

Appendix V: Equipment List/Sizing

Table 8: List of Equipment for Pretreatment of Sugarcane Bagasse (Based on Processes done on Aspen)

Unit	Equipment Type	Quantity	Size/Power	Material
CU-101	Crusher/Grinder	1	153 T/hr	Mild Steel
AH-101	Reactor	1	Diameter: 1.83m Height: 6.71m	Carbon Steel
CF-101	Centrifuge	1	Diameter: 1.28m	Stainless Steel

Table 9: List of Equipment for Fermentation Process

Unit	Equipment Type	Quantity	Size/Power	Material
CF-102	Centrifuge	1	Diameter: 1.28m	Stainless Steel
MX-101	Mixer	1	NA	Carbon Steel
MX-102	Mixer	1	NA	Carbon Steel
FR-101 FR-102 FR-103	Fermenters	3	Diameter: 1.52m Height: 5.94m	Carbon Steel
CE-101	Chemostat	1	Diameter: 2.44m Height: 4.42m	Stainless Steel
AH-102	Reactor	1	Diameter: 1.83m Height: 6.71m	Carbon Steel
F-101	Membrane Filter	1	NA	Carbon Steel

Table 10: List of Equipment for Purification Process

Unit	Equipment Type	Quantity	Size/Power	Material
FL-101	Flash Drum	1	Diameter: 2.44m Height: 4.42m	Carbon Steel
P-101	Centrifugal Pump	1	Efficiency: 0.75 Head: 10.82m	Carbon Steel
P-102	Centrifugal Pump	1	Efficiency: 0.54 Head: 0.18m	Carbon Steel
P-103	Centrifugal Pump	1	Efficiency: 0.54 Head: 0.18m	Carbon Steel
D-101	Distillation Column	1	Diameter: 5.8m Height: 14.63m #Trays: 26	Carbon Steel
D-102	Distillation Column	1	Diameter: 0.59m Height: 5.49m #Trays: 11	Carbon Steel
D-103	Distillation Column	1	Diameter: 1.02m Height: 12.19m #Trays: 22	Carbon Steel
MX-103	Mixer	1	NA	Carbon Steel
RB-101	Kettle Reboiler	1	Heat transfer area: $3166m^2$	Carbon Steel
RB-102	Kettle Reboiler	1	Heat transfer area: $48m^2$	Carbon Steel
RB-103	Kettle Reboiler	1	Heat transfer area: $8m^2$	Carbon Steel
C-101	Partial Condenser	1	Diameter: 2.3m Height: 6.4m	Carbon Steel
C-102	Partial Condenser	1	Diameter: 0.91m Height: 2.74m	Carbon Steel
C-103	Partial Condenser	1	Diameter: 0.91m Height: 2.74m	Carbon Steel

Table 11: List of Equipment for Polymerization Process (Based on averaged specs from purification/fermentation equipment)

Unit	Equipment Type	Quantity	Size/Power	Material
R-101 R-102 R-103	Reactor	3	Diameter: 1.83m Height: 6.71m	Carbon Steel
P-104	Centrifugal Pump	1	Efficiency: 0.75 Head: 10.82m	Carbon Steel
P-105	Centrifugal Pump	1	Efficiency: 0.75 Head: 10.82m	Carbon Steel
D-104	Distillation Column	1	Diameter: 2.47m Height: 10.77m #Trays: 20	Carbon Steel
MX-104	Mixer	1	NA	Carbon Steel
D-105	Distillation Column	1	Diameter: 2.47m Height: 10.77m #Trays: 20	Carbon Steel
N-101	Neutralizing Vessel	1	Diameter: 2.44m Height: 4.42m	Carbon Steel
EV-101	Evaporator	1	Diameter: 2.44m Height: 4.42m	Carbon Steel

Appendix VI: Sample Calculation for Equipment Sizing

The following is a sample hand calculation for the sizing of the distillation column, D-103, which separates methanol and water to be recycled back into the process. The flowrates of the feed, distillate, and bottom and the reflux ratio and boil up ratios were taken from Aspen. Figure 8, below, shows the distillation column results which are used to calculate the operating lines for the McCabe Thiele Diagram in order to find the number of trays in the column.

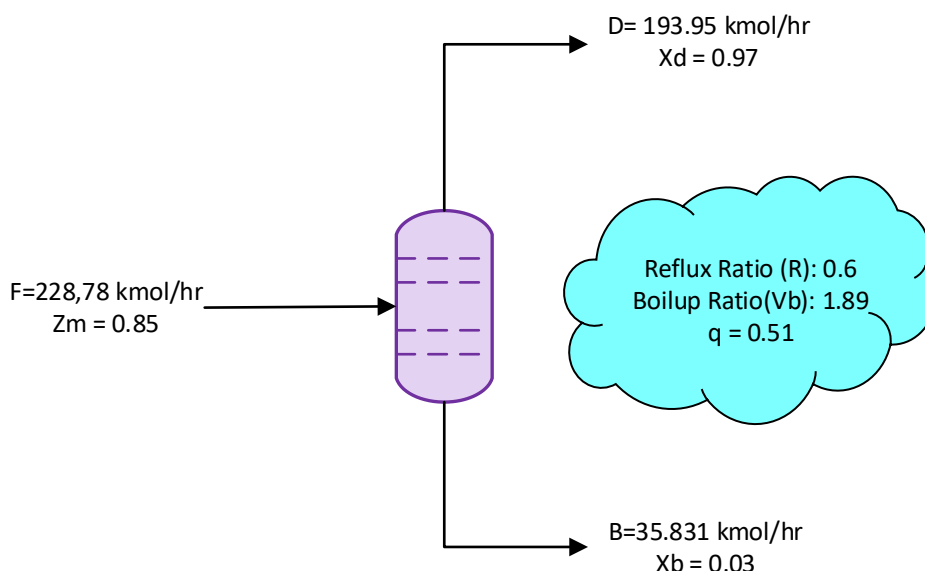


Figure 17: Distillation Column Information Used for Hand Calculations

The McCabe Thiele Diagram consists of an equilibrium curve, $y=x$ curve, a top operating line, bottom operating line, and a feed line. The equilibrium curve was found using the Engineering Toolbox website for methanol and water. In order to find the top operating line, the following equation was used:

$$y = \frac{L}{V}x + \frac{D}{V}x_D \quad (1)$$

Where

$$\frac{L}{V} = \frac{R}{R+1} = \frac{0.6}{0.6+1} = 0.38$$

The reflux rate was also found on Aspen to be 116.37 kmol/hr, so using the $\frac{L}{V} = 0.38$, V was solved to be 310.32 kmol/hr. Therefore, the top operating line equation came out to be:

$$y = 0.38x + 0.61 \quad \text{(Top Operating Line)}$$

To find the bottom operating line, the following equation was used:

$$y = \frac{\bar{L}}{\bar{V}}x - \frac{B}{\bar{V}}x_B \quad (2)$$

Where

$$\frac{\bar{L}}{\bar{V}} = \frac{V_B + 1}{V_B} = \frac{1.89 + 1}{1.89} = 1.53$$

The boil-up rate was found on Aspen to be 67.87 kmol/hr and used the equation above to solve for the missing variable and formulated the bottom operating line to be:

$$y = 1.53x - 0.0158 \quad \text{(Bottom Operating Line)}$$

The last operating line to be found is the feed line, which is found by the following equation:

$$y = \frac{q}{q-1}x + \frac{z_F}{1-q} \quad (3)$$

Using this equation, the feed line equation is:

$$y = -1.04x + 1.73 \quad \text{(Feed Operating Line)}$$

These operating equations, along with the equilibrium curve and the $y=x$ line, are drawn into the McCabe Thiele Diagram shown below. The number of stages drawn are 19 stages, therefore there are 19 trays inside the distillation column.

$$N = 19 \text{ trays}$$

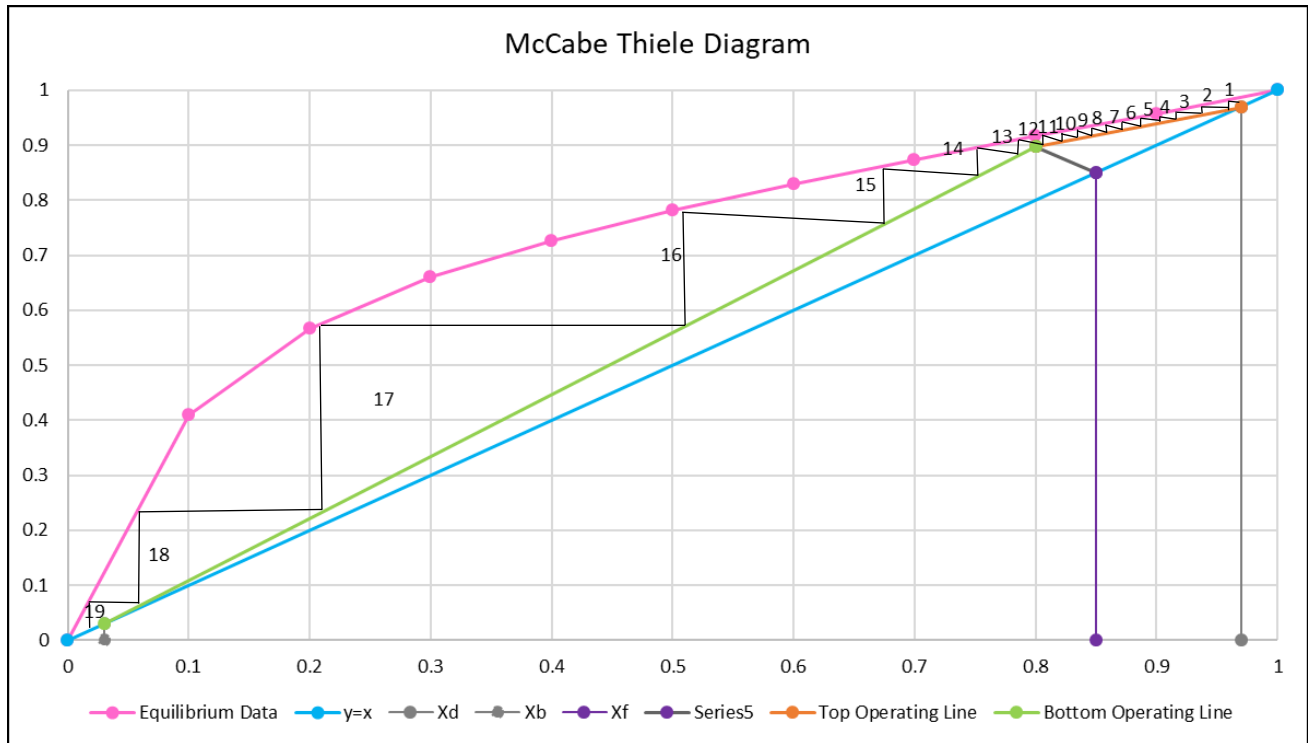


Figure 18: McCabe Thiele Diagram for Number of Stages

Now, in order to determine the height of the distillation column the number of trays (N) which was calculated to be 19 trays needs to be multiplied by the plate spacing (l_t) which is found on Aspen to be 0.61 meters.

$$D_h = N * l_t \quad (4)$$

$$D_h = 11.59 \text{ meters}$$

Next, the diameter of the column is calculated through a series of equations as shown below. Equation 5 determines the maximum permissible velocity in the column.

$$U_{perm} = k \sqrt{\frac{\rho_l - \rho_v}{\rho_v}} \quad (5)$$

Where

k = sizing parameter, 0.021 m/s found through literature^[26]

ρ_l = liquid density, $860.103 \frac{kg}{m^3}$ found through Aspen

ρ_v = vapor density, $1.135 \frac{kg}{m^3}$ found through Aspen

$$U_{perm} = 0.577 \frac{m}{s}$$

The maximum permissible velocity is then plugged into Equation 6 to determine the cross-sectional area of the column.

$$A_c = \frac{V \times MW_v}{U_{perm} \times 3600 \times \rho_v} \quad (6)$$

Where

V = vapor flow rate, $67.87 \frac{kmol}{kg}$ found through Aspen

MW_v = molecular weight of methanol, $32.04 \frac{kg}{kmol}$

U_{perm} = maximum permissible velocity, $0.577 \frac{m}{s}$

ρ_v = vapor density, $1.135 \frac{kg}{m^3}$ found through Aspen

$$A_c = 0.921 m^2$$

The cross-sectional area is then plugged into Equation 7 which gives the diameter of the column.

$$D_c = \sqrt{\frac{4A_c}{\pi}} \quad (7)$$

Where

A_c = cross-sectional area, $0.921 m^2$

$$D_c = 1.083 \text{ meters}$$

Analysis of Sample Calculations and Aspen Simulation Calculations

Overall, the hand calculated values for different parts of the distillation column did not stray far from the values determined through Aspen. The number of trays and the feed stage values were determined through the McCabe Thiele diagram and because of the nature of having to hand draw the stages, there will almost always be a discrepancy especially when the stages drawn are very small. The McCabe Thiele determined the column to have 19 trays with a feed stage at the 12th tray, while Aspen determined there to be 22 trays with a feed stage at the 14th tray. The height of the column was calculated to be 11.59 meters, while Aspen determined the column of the height to be 12.19 meters. The reason for this discrepancy is because the hand calculated height uses the hand calculated number of trays rather than the number of trays found through Aspen. Lastly, the diameter of the column was calculated to be 0.183 meters through a series of equations, but since the sizing parameter used in those equations was found in literature it may not accurately depict the sizing parameter for the actual column. In Aspen, the diameter of the column was determined to be 1.20 meters. The table below is a compilation of hand calculated values compared to the values determined by Aspen.

Table 12: Hand calculated Values vs. Aspen Values for D-103

Description	Hand Calculated Values	Aspen Values	Percent Error
Number of trays	19 trays	22 trays	13.63%
Feed stage	Tray 12	Tray 14	14.28%
Column Height	11.59 meters	12.19 meters	4.92%
Column Diameter	1.183 meters	1.20 meters	1.41

Appendix VII: Preliminary Utilities

Table 13: Preliminary Utilities for Fermentation and Purification Process from ASPEN

Description	Type	Units	PFD Name	Equipment Type	Rate
Electricity (U1)	Power	kW	P-101	Pump	3.4251
	Power	kW	P102	Pump	0.020789
	Power	kW	AH-102	Reactor	46151
Cooling Water (U2)	Condenser	kg/hr	D-101	Condenser	2608226
	Condenser	kg/hr	D-102	Condenser	133473
	Condenser	kg/hr	D-103	Condenser	200502
Low Pressure Steam (U3)	Utility usage	kW	EST	Reactor Distillation	440.89
	Utility usage	kW	FR-10X	Reactor	59790
	Utility usage	kW	FL-101	Flush Reactor	68101
Medium Pressure Steam (U4)	Reboiler usage	kg/hr	D-101	Distillation Reboiler	106515
	Reboiler usage	kg/hr	D-103	Distillation Reboiler	1304.7
	Utility usage	kW	WAHY	Reactor Distillation	5184.4
High pressure Steam (U5)	Reboiler utility usage	kg/hr	D-102	Distillation Reboiler	3433.4

Table 14: Preliminary Utilities for Polymerization using Rough Estimations and Hand Calculations

Description	Type	Units	PFD Name	Equipment Type	Rate
Electricity (U1)	Power	kW	R-101	Reactor	46151
Cooling Water (U2)	Condenser	kg/hr	D-104	Condenser	2608226
	Condenser	kg/hr	D-105	Condenser	133473
Low Pressure Steam (U3)	Utility usage	kW	R-102	Reactor	440.89
	Utility usage	kW	EV-101	EV	68101
Medium Pressure Steam (U4)	Reboiler usage	kg/hr	D-104	Distillation Reboiler	106515
	Reboiler usage	kg/hr	D-105	Distillation Reboiler	1304.7
	Utility usage	kW	R-103	Reactor	5184.4